

Carbon Audit & Scientific Verification Brief: Unknown Site

1. Site Metadata & Governance

- **Site / District Name:** Unknown Site
- **Audit Date:** 2026-08-23 14:46
- **Geometry Type:** Polygon
- **Spatial Area:** 250.00 Hectares
- **National Jurisdiction:** Republic of Rwanda (WDPA Protected Buffer Zone)

2. Zonal Carbon Analysis & Audit Metrics

- **Mean Above-Ground Biomass (AGB):** 0.0 Mg/ha
- **Historical AGB Trend:** +0.50 Mg/ha/yr
- **Total Carbon Stock:** 0 Mg C
- **CO2 Equivalent (CO2e) Stock:** 0 Mg CO2e
- **Estimated Offset Valuation (@ \$10/tCO2e):** \$0.00 Million USD
- **Net Carbon Flux Status:**

3. Scientific Interpretation & Recommendations

AGB Analysis: Critical Degradation / Severe Soil Exposure

- **Meaning:** Values below 15 Mg/ha indicate severe biomass loss, likely from clear-cutting or intense agricultural degradation.
- **Recommendation:** Immediate radical terracing & pioneer planting (e.g., Acacia polyacantha) required to stabilize soil.

Sequestration Trend: Healthy Active Sequestration

- **Meaning:** Positive flux up to 0.8 Mg/ha/yr represents steady, verifiable carbon drawdown.
- **Action Required:** Standard high-yield carbon credit asset. Continue current monitoring.

4. Historical Satellite Biomass & CO2e Timeseries (2010 - 2022)

Year	Mean AGB (Mg/ha)	Total Carbon Stock (Mg C)	CO2e Potential (Mg)
2022	23.1 Mg/ha	5,775 Mg	7,653,829 Mg

5. AI 10-Year Growth & Degradation Risk Prediction

- **Model Architecture:** Random Forest ML Regressor (Trained on 2010-2020 ESA CCI Data)
- **Model Validation Metrics:** R-squared = 0.74 | RMSE = 14.2 Mg/ha (Validated against NASA GEDI LiDAR)
- **10-Year Projected Growth:** +5.0 Mg/ha forecast by 2032
- **Risk Assessment:** Low Degradation Risk - Ecosystem actively accumulating carbon.

6. Recommended Native Rwandan Tree Species (Restor.eco Standard)

- **Markhamia lutea (Umusave)** - Fast-growing agroforestry hardwood (Density = 0.54 g/cm3)
- **Polyscias fulva (Umurava)** - High-biomass montane canopy tree (Density = 0.42 g/cm3)
- **Acacia polyacantha (Umugondo)** - Nitrogen-fixing soil builder (Density = 0.68 g/cm3)
- **Podocarpus latifolius (Umuseke)** - Climax cloud forest giant (Density = 0.56 g/cm3)

7. Scientific Methodology & Dataset Provenance

[RW] LCRI Scientific Verification & Carbon Audit Report

All calculations conform to the **IPCC AR6 Guidelines for National Greenhouse Gas Inventories** and the **Chave et al. (2014)** tropical allometric equations using **ESA CCI Biomass v7.0** raster data.